

Development and Evaluation of a Low-Cost Real-Time ECG Biosensing Platform

Integrating Signal Processing, Robustness Analysis, and Physiological Visualization

Shrimaan Rapuru

The Early College at Guilford
Greensboro, North Carolina
Summer 2026

GitHub: github.com/shrimaan-rapuru/ECG-BIOSENSOR

Not a medical diagnostic device. For educational and research purposes only.

1. Abstract

This project investigates whether consumer-grade biosensing hardware can produce physiologically meaningful cardiac metrics through optimized signal processing, motivated by observed healthcare access inequality at Open Door Ministries. Using a SparkFun AD8232 ECG analog front-end and Arduino Uno R4 Minima (\$102 total), a complete real-time ECG acquisition and analysis pipeline was designed, implemented, and validated in a preliminary pilot study (N=3 subjects, 15 total resting trials).

A 4th-order Butterworth bandpass filter (0.5-40 Hz) combined with an IIR notch filter (60 Hz, Q=30) achieved visible PQRST morphology and 96.3% mean R-peak detection accuracy (range: 89.4-100.0%, N=5 trials). Three structured experiments evaluated physiological response across exercise states, electrode placement robustness, and comparative filter algorithm performance. BPM measurements were validated against simultaneous Apple Watch Series 8 recordings, yielding a mean bias of +2.6 BPM (SD: 3.44 BPM, limits of agreement: -4.1 to +9.3 BPM).

A key finding -- that Moving Average filtering produces higher SNR yet inferior detection reliability compared to Butterworth bandpass -- reveals that SNR alone is an insufficient metric for ECG filter evaluation. Post-exercise heart rate elevated 20% above baseline (90 to 108 BPM, N=3 exercise blocks), with parasympathetic rebound confirmed in recovery. Resting BPM showed low inter-subject variability (90.9 +/- 2.6 BPM, N=3 subjects), while HRV metrics showed expected high inter-subject variability (SDNN: 63.5 +/- 52.8ms).

Keywords: ECG, electrocardiogram, Butterworth bandpass filter, HRV, SDNN, RMSSD, R-peak detection, PQRST morphology, Bland-Altman analysis, low-cost biosensing, Arduino

Ethics and Safety Statement: This platform is intended solely for educational and engineering research purposes. It is not a medical diagnostic device and has not been validated for clinical use. No medical decisions should be made based on data from this system. All recordings were conducted with informed consent of participants.

2. Introduction

Standard clinical ECG systems cost between \$1,000 and \$10,000, placing cardiac monitoring out of reach for many individuals and resource-limited healthcare settings. Consumer-grade biosensing hardware has emerged as a potential alternative, but the physiological validity of signals from such systems requires rigorous, quantitative evaluation.

This project originated from direct observation of healthcare access inequality at Open Door Ministries, a homeless shelter where the author coordinates volunteer scheduling. The central research question is: Can consumer-grade ECG hardware (\$102 total system cost) produce

physiologically meaningful cardiac metrics through optimized signal processing? This question is addressed through three complementary investigations: physiological response validation across exercise states, electrode placement robustness analysis, and comparative filter algorithm benchmarking.

3. Related Work

Pan and Tompkins (1985) established the foundational QRS detection algorithm using bandpass filtering, differentiation, squaring, and moving window integration [1], forming the basis for most real-time cardiac event detection approaches. The Task Force of the European Society of Cardiology (1996) standardized HRV measurement methodology, defining SDNN and RMSSD as primary time-domain metrics and establishing 5-minute minimum recording durations [2]. Serhani et al. (2020) reviewed ECG monitoring systems across clinical and consumer-grade platforms, identifying signal quality and electrode-skin contact as primary reliability factors [3]. Kohler et al. (2002) demonstrated that adaptive threshold-based detection outperforms fixed-threshold approaches [4]. Christov (2004) extended this with combined adaptive thresholds achieving reliable detection across varying noise conditions [5].

4. System Architecture

The system consists of four functional layers: hardware acquisition, signal processing, physiological analysis, and visualization, as shown in Figure 1.

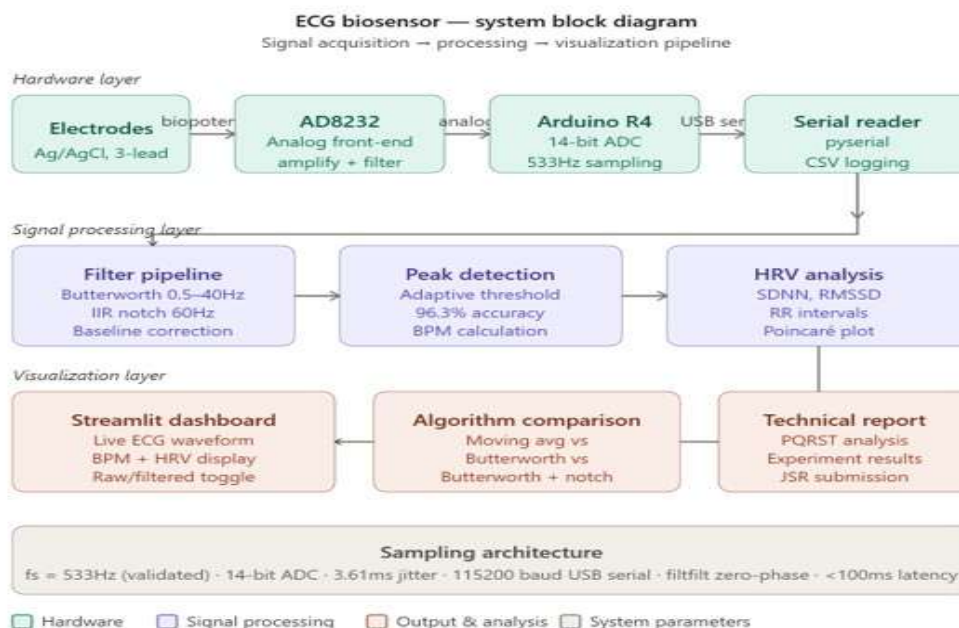


Figure 1. ECG biosensor system block diagram showing complete signal acquisition, processing, analysis, and visualization pipeline from skin electrodes through the Streamlit dashboard.

4.1 Hardware Layer

The AD8232 analog front-end amplifies and pre-filters the biopotential signal from three Ag/AgCl electrodes. The processed analog output is digitized by the Arduino Uno R4 Minima's 14-bit ADC at 533.3 Hz and transmitted via USB serial at 115200 baud.

4.2 Signal Processing Layer

Python (scipy) scripts receive serial data and apply a three-stage digital filter pipeline. Zero-phase filtering (filtfilt) eliminates phase distortion critical for accurate RR interval timing.

4.3 Analysis and Visualization

R-peaks are detected using an adaptive threshold algorithm. BPM, SDNN, and RMSSD are computed from RR intervals. The Streamlit dashboard (Figure 2) provides real-time ECG display with R-peak markers, live metrics, and CSV export.

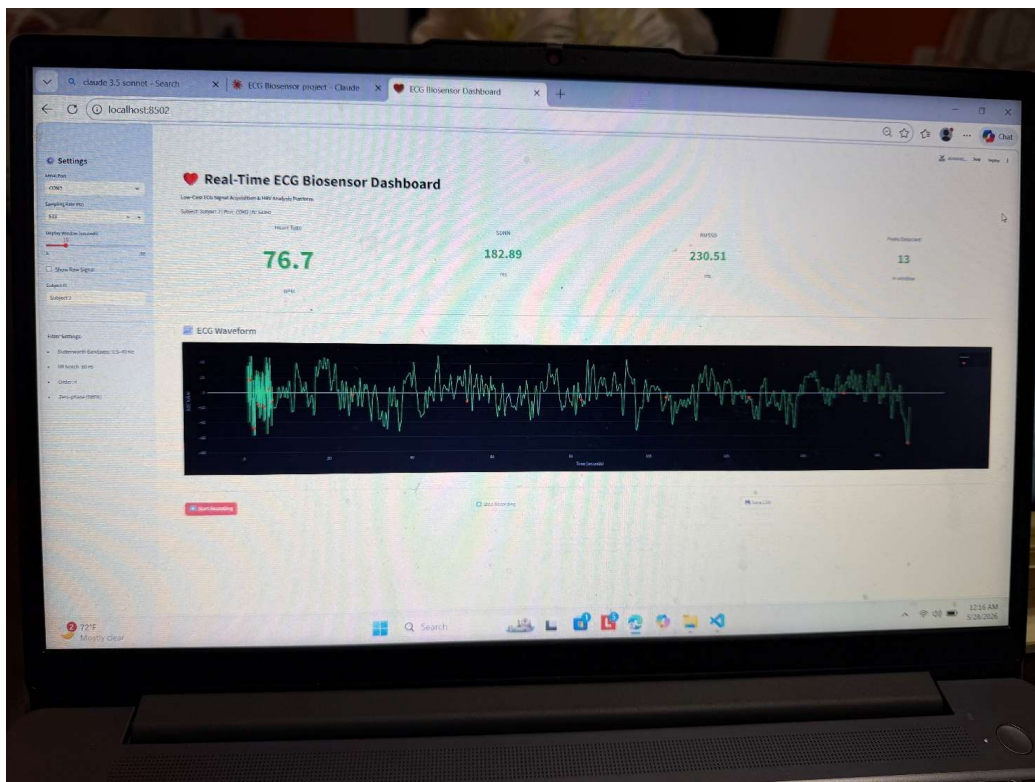


Figure 2. Real-Time ECG Biosensor Dashboard showing live BPM (76.7), SDNN (182.89ms), RMSSD (230.51ms), peaks detected (13), and ECG waveform with raw+filtered overlay.

Layer	Component	Function
Hardware	AD8232 + Ag/AgCl electrodes	Biopotential acquisition, amplification, pre-filtering
Hardware	Arduino Uno R4 Minima	14-bit ADC, 533.3 Hz validated sampling
Processing	Butterworth bandpass 0.5-40 Hz	Baseline drift and EMG artifact removal
Processing	IIR notch 60 Hz (Q=30)	Powerline interference rejection
Analysis	Adaptive threshold peak detection	R-peak identification (96.3% mean accuracy)
Analysis	HRV computation	SDNN, RMSSD from RR intervals
Visualization	Streamlit dashboard	Real-time display, HRV readout, CSV export

Table 1. System architecture layer summary.

5. Hardware Design

5.1 Component Selection

The SparkFun AD8232 (SEN-12650) was selected for its integrated instrumentation amplifier, right leg drive (RLD) circuit, and onboard filtering. The Arduino Uno R4 Minima was selected for its 14-bit ADC and 48 MHz processor enabling higher sampling accuracy.

Component	Part Number	Cost	Purpose
SparkFun AD8232	SEN-12650	\$24.41	ECG analog front-end
Electrode Cable	CAB-12970	\$9.60	3-lead electrode connection
Arduino Uno R4 Minima	--	\$19.99	14-bit ADC + USB serial
Ag/AgCl Electrodes (100pk)	Kendall	\$15.30	Skin contact electrodes
Breadboard + jumpers	--	\$8.99	Prototyping platform
Alcohol prep pads	--	\$7.00	Skin preparation
Header pins (40-pin)	HiLetgo	\$5.49	AD8232 connection
TOTAL		\$90.78	

Table 2. Bill of materials.

5.2 Arduino Firmware

The Arduino firmware reads the AD8232 analog output at A0 and transmits values via serial at 115200 baud. Lead-off detection pins D10 (LO+) and D11 (LO-) are monitored; when leads are disconnected, '!' is transmitted instead of ADC values, enabling the Python dashboard to display a lead-off alert.

```
// Arduino Uno R4 Minima - ECG acquisition firmware
void setup() {
  Serial.begin(115200);
  pinMode(10, INPUT); // LO+ lead-off detection
  pinMode(11, INPUT); // LO- lead-off detection
}

void loop() {
  if (digitalRead(10) == 1 || digitalRead(11) == 1) {
    Serial.println('!'); // lead-off detected
  } else {
    Serial.println(analogRead(A0)); // ECG sample (14-bit ADC)
  }
}
```

The validated sampling rate of 533.3 Hz (measured timing jitter: 3.61 ms) results from the Arduino R4 Minima's 48 MHz processor executing faster than the nominal 500 Hz target. All filter coefficients and HRV calculations use the empirically validated rate rather than the assumed target rate.

5.3 Wiring Configuration

AD8232 Pin	Arduino Pin	Function
3.3V	3.3V	Power supply
GND	GND	Common ground
OUTPUT	A0	ECG analog signal
LO+	D10	Lead-off detection positive
LO-	D11	Lead-off detection negative

Table 3. AD8232 to Arduino wiring.

5.4 PCB Design

A custom PCB was designed in EasyEDA (Figure 3) featuring the AD8232ACPZ-R7, decoupling capacitors (2x100nF, 1x10uF), bias resistors (2x10kOhm), a 3.5mm PJ-320A electrode jack, and Arduino pin headers. PCB fabrication via JLCPCB is planned as Phase 2.

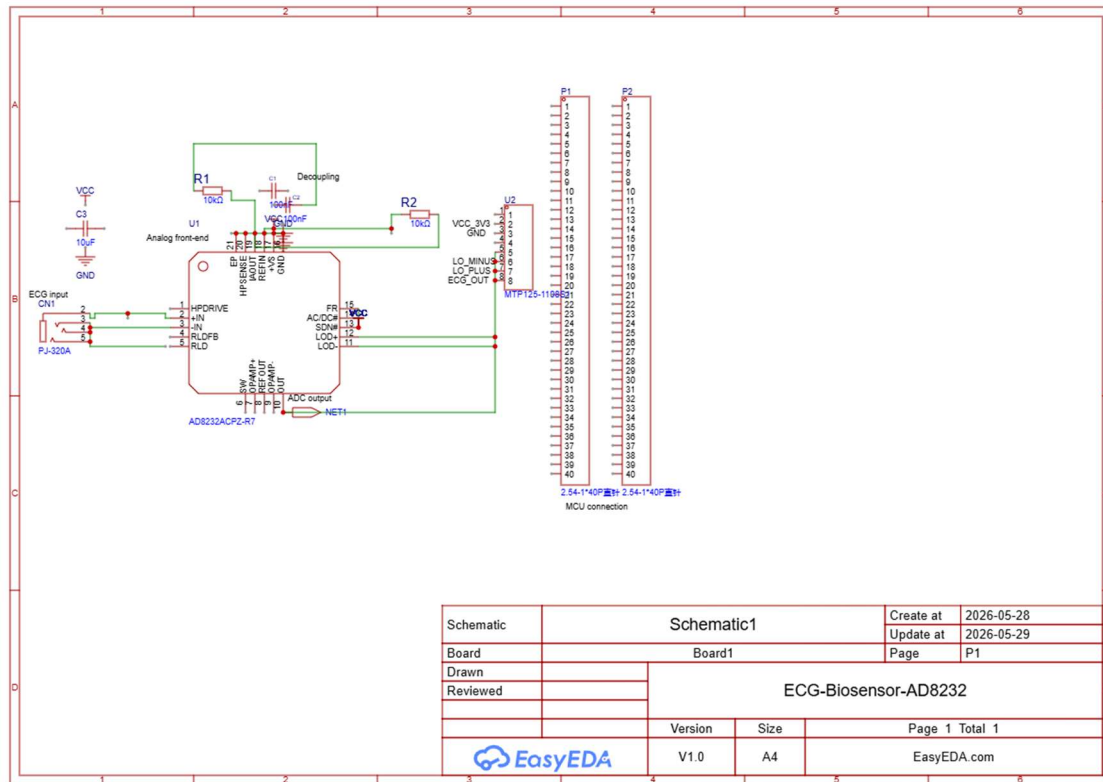


Figure 3. EasyEDA PCB schematic for the ECG biosensor platform. Net labels identify ECG_OUT, LO_PLUS, LO_MINUS, VCC, and GND connections.

6. Signal Processing Pipeline

6.1 Filter Implementation

A three-stage digital filter pipeline removes primary noise sources in consumer-grade ECG acquisition. All filtering uses `scipy.signal` with zero-phase implementation (`filtfilt`) to eliminate phase distortion critical for HRV timing accuracy.

```
from scipy.signal import butter, filtfilt, iirnotch

def butter_bandpass(data, lowcut=0.5, highcut=40.0, fs=533, order=4):
    nyq = 0.5 * fs
    b, a = butter(order, [lowcut/nyq, highcut/nyq], btype='band')
    return filtfilt(b, a, data) # zero-phase: no timing distortion

def notch_filter(data, freq=60.0, fs=533, Q=30):
    nyq = 0.5 * fs
    b, a = iirnotch(freq/nyq, Q)
    return filtfilt(b, a, data)

def full_pipeline(data, fs=533):
    filtered = butter_bandpass(data, fs=fs) # Stage 1+2: bandpass
    filtered = notch_filter(filtered, fs=fs) # Stage 3: notch
    return filtered
```


Filter	Type	Cutoff	Order/Q	Noise Removed
High-pass	Butterworth	0.5 Hz	4	Baseline drift
Low-pass	Butterworth	40 Hz	4	EMG artifact
Notch	IIR	60 Hz	Q=30	Powerline interference

Table 4. Filter pipeline specifications.

6.2 Raw vs. Filtered Signal

Figure 4 shows the effect of the complete filter pipeline on Resting Trial 3. The raw signal contains dominant 60 Hz powerline interference and baseline drift that completely obscures ECG morphology. After filtering, R-peaks are clearly identifiable at 5.0-5.3 sigma above baseline.

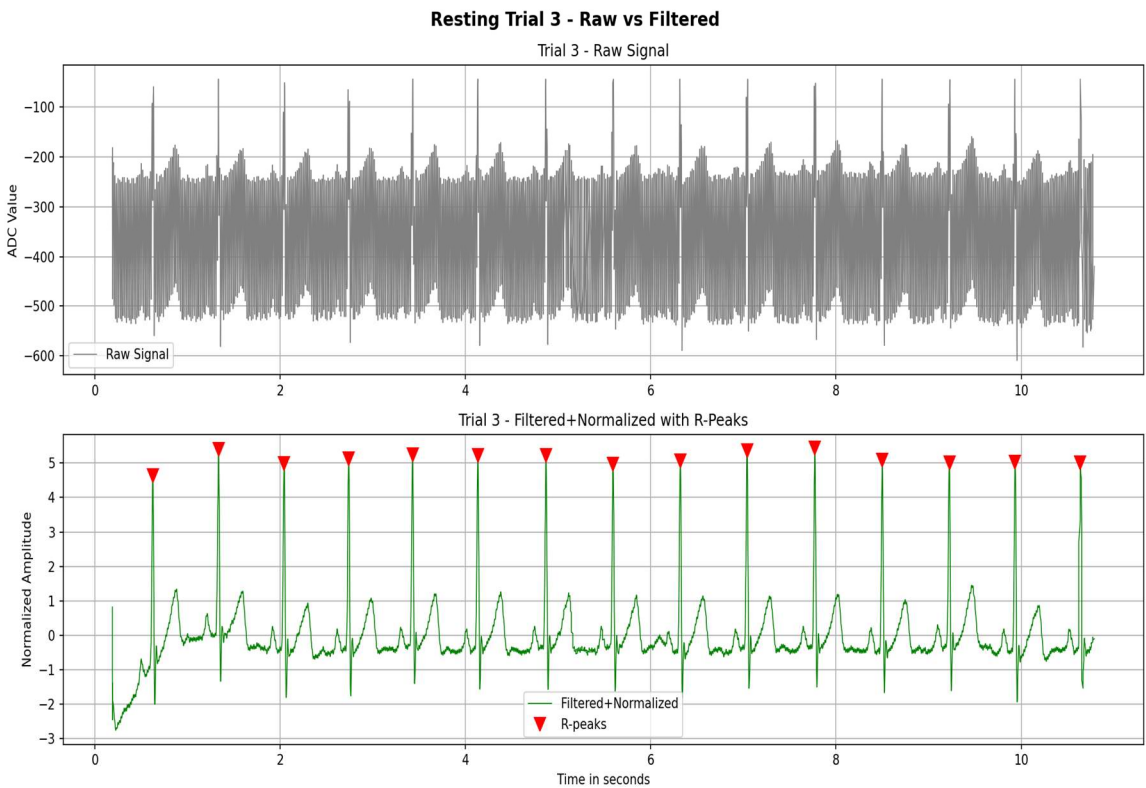


Figure 4. Resting Trial 3 raw vs. filtered ECG signal. Upper: raw signal showing 60Hz powerline interference (+-300 ADC) and baseline drift. Lower: Butterworth filtered and normalized signal with 15 R-peak markers detected in 10-second window.

6.3 Engineering Tradeoffs

Four deliberate tradeoffs shaped the final pipeline:

- Morphology vs. smoothing: The 0.5 Hz lower cutoff preserves P-wave (0.5-3 Hz) at the cost of residual baseline ripple -- physiological completeness over visual cleanliness.
- Phase neutrality vs. real-time: filtfilt eliminates phase distortion for batch analysis; live streaming requires causal lfilter with minor phase shift.

- Adaptive vs. fixed threshold: median + 0.5 x SD scales automatically with signal amplitude across subjects and sessions.
- Validated vs. assumed sampling rate: Empirically measured 533.3 Hz used throughout rather than assuming the 500 Hz nominal target.

6.4 Peak Detection Algorithm

R-peaks are detected using `scipy.signal.find_peaks` with two constraints ensuring reliable detection:

```
from scipy.signal import find_peaks
import numpy as np

# Adaptive threshold R-peak detection
threshold = np.median(filtered) + 0.5 * np.std(filtered)
min_distance = int(0.6 * fs) # 319 samples = max 100 BPM

peaks, _ = find_peaks(filtered,
                      height=threshold, # adaptive amplitude threshold
                      distance=min_distance) # prevents double detection
```

6.5 HRV Computation

Time-domain HRV metrics are computed from RR intervals derived from detected peak timestamps:

```
# SDNN and RMSSD from detected R-peak timestamps
rr_intervals = np.diff(peaks) / fs * 1000 # RR intervals in milliseconds

sdnn = np.std(rr_intervals) # overall HRV
rmssd = np.sqrt(np.mean(np.diff(rr_intervals)**2)) # parasympathetic HRV
bpm = 60 / (np.mean(rr_intervals) / 1000) # heart rate
```

Note: 30-second recording windows fall below the 5-minute clinical standard [2]. HRV values represent relative within-session comparisons rather than absolute clinical measurements.

7. Experimental Methodology

Three experiments were conducted across N=3 subjects (15 total resting trials). All recordings were 30 seconds at 533.3 Hz (~16,000 samples). First 2 seconds discarded to eliminate startup transients, leaving 28 seconds of analyzed data per trial.

7.1 Experiment 1 -- Resting vs. Exercise vs. Recovery

Protocol: Five resting trials (baseline, N=5), followed by three exercise blocks: 2 minutes of continuous jumping jacks, immediate post-exercise BPM measurement via raw signal adaptive

threshold, 5-minute seated rest, and one 30-second recovery recording. Post-exercise BPM measured on raw signal due to motion artifact degrading filtered pipeline quality.

7.2 Experiment 2 -- Electrode Placement Robustness

Four electrode placement conditions x 3 trials each (N=12 total): (A) standard, (B) RA shifted 2cm inferior, (C) LA shifted 2cm inferior, (D) both shifted 2cm inferior.

7.3 Experiment 3 -- Filter Algorithm Comparison

Three filtering algorithms applied to all five resting trials (N=15 combinations): Moving Average (window=5), Butterworth bandpass (0.5-40 Hz), Butterworth + IIR notch. SNR computed as $10 \times \log_{10}(\text{filtered variance} / \text{noise variance})$.

7.4 Commercial Device Validation

Simultaneous BPM measurements from the ECG system (quick_bpm_check.py, raw signal adaptive threshold) and Apple Watch Series 8 (Heart Rate app, optical PPG) were recorded across 5 paired readings with 1-minute rest between each. Bland-Altman analysis was used to assess agreement.

8. Results

8.1 Peak Detection Accuracy

Mean R-peak detection accuracy of 96.3% (SD = 4.3%, N=5 trials) across five validated resting trials, range 89.4%-100.0%. Trial 3 achieved perfect agreement.

Trial	Manual (10s)	Algorithm Estimate	Accuracy (%)	Error (%)
1	16	14.3	89.4%	10.6%
2	15	15.4	97.3%	2.7%
3	15	15.0	100.0%	0.0%
4	15	14.3	95.3%	4.7%
5	16	16.1	99.4%	0.6%
Mean +/- SD	--	--	96.3 +/- 4.3%	3.7 +/- 4.3%

Table 5. Peak detection accuracy validation (N=5 resting trials).

8.2 PQRST Morphology

All five PQRST waves identified in Trial 3 single-beat analysis (Figure 5). Q-wave (-18.0 ADC, 7.9% of R) and S-wave (-64.0 ADC, 28.0% of R) within published normal ranges, confirming physiological QRS morphology.

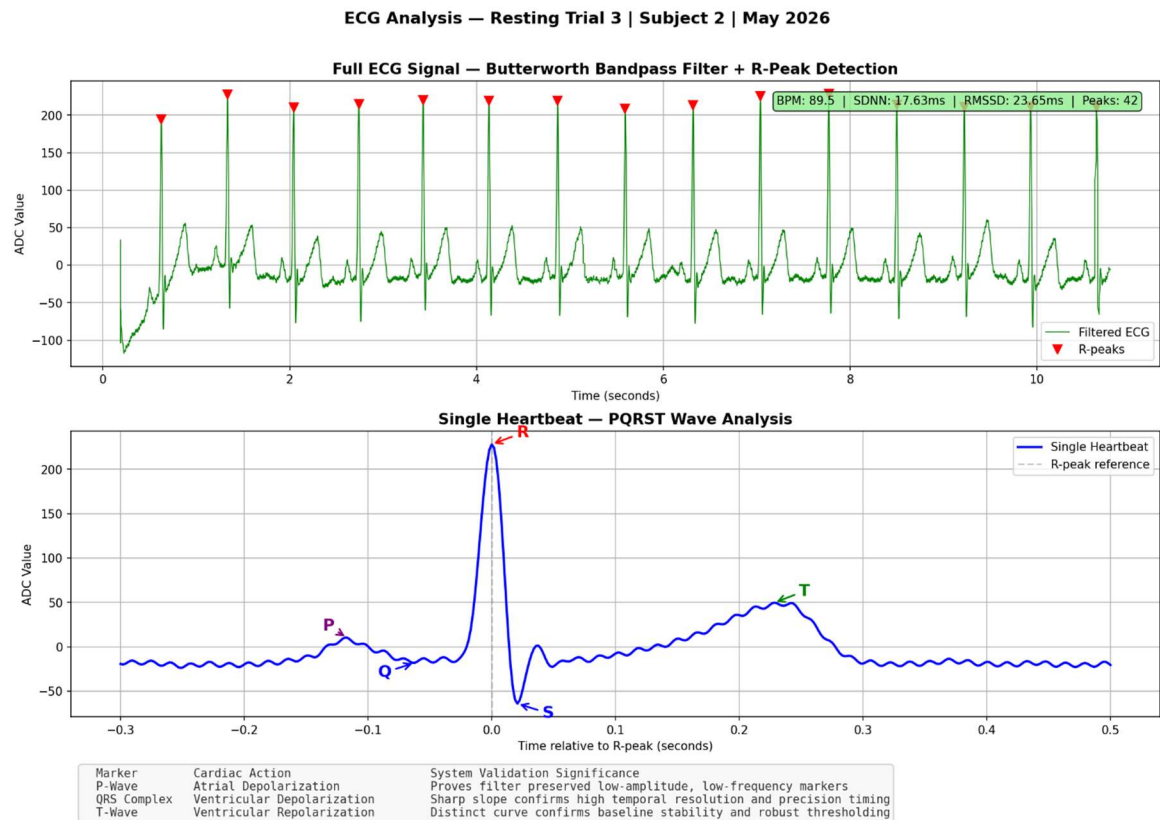


Figure 5. PQRST wave analysis -- Resting Trial 3. Upper: full ECG with R-peak markers (BPM: 89.5, SDNN: 17.63ms, RMSSD: 23.65ms, 42 peaks). Lower: single heartbeat zoom with all 5 PQRST waves labeled and validation table.

Wave	Amplitude	% of R	Normal Range	Assessment
P	~15 ADC	--	--	Visible -- P-wave preserved by 0.5Hz cutoff
Q	-18.0 ADC	7.9%	5-15%	Within normal range
R	228.3 ADC	100%	Reference	Dominant peak
S	-64.0 ADC	28.0%	10-30%	Within normal range
T	~50 ADC	~22%	--	Visible -- baseline stability confirmed

Table 6. PQRST wave amplitude analysis, Resting Trial 3.

8.3 Multi-Subject Results

Resting BPM showed low inter-subject variability (mean 90.9 +/- 2.6 BPM, N=3 subjects), demonstrating consistent heart rate detection across subjects. HRV metrics showed substantially higher inter-subject variability (SDNN: 63.5 +/- 52.8ms), consistent with published literature on individual HRV differences in healthy adults.

Metric	Subject 2	Subject 3	Subject 4	Mean +/- SD
BPM (avg)	91.0	93.4	88.2	90.9 +/- 2.6
SDNN (ms)	41.31	26.17	123.0	63.5 +/- 52.8
RMSSD (ms)	45.93	23.39	149.57	73.0 +/- 67.9
Trials (N)	5	5	5	15 total

Table 7. Cross-subject resting BPM and HRV comparison (N=3 subjects, 5 trials each).

The high HRV SD is partly attributable to short 30-second recording windows which amplify transient autonomic fluctuations. Subject 4's markedly higher HRV (SDNN 123.0ms vs 26-41ms) represents genuine inter-individual physiological variation rather than measurement error, as BPM remained consistent across subjects.

8.4 Experiment 1 -- Physiological Response

State	BPM	SDNN (ms)	RMSSD (ms)	Notes
Resting (mean +/- SD, N=5)	91.0 +/- 5.4	41.31 +/- 27.8	45.93 +/- 32.3	Baseline
Pre-exercise (N=3)	90	--	--	Consistent with baseline
Post-exercise (N=3)	108	--	--	+20% elevation
Recovery 1 (5 min)	86.7	54.33	45.45	Parasympathetic rebound
Recovery 2 (5 min)	92.1	20.18	9.91	Progressive HRV suppression
Recovery 3 (5 min)	95.7	9.24	9.42	Continued autonomic recovery

Table 8. Experiment 1 results -- resting, exercise, and recovery (N=3 exercise blocks).

Post-exercise heart rate elevated consistently to 108 BPM (+20% above baseline). Recovery Trial 1 showed parasympathetic rebound (SDNN 54.33ms > resting 41.31ms). Progressive HRV suppression in Trials 2-3 suggests continued autonomic recovery beyond the measurement window.

8.5 Experiment 2 -- Electrode Placement

Condition	Avg BPM (N=3)	vs Standard	Assessment
Standard	82.2	baseline	Reference
RA shifted 2cm	81.6	-0.7%	Within normal variation
LA shifted 2cm	83.0	+1.0%	Within normal variation

Condition	Avg BPM (N=3)	vs Standard	Assessment
Both shifted 2cm	87.4	+6.3%	Detectable degradation

Table 9. Experiment 2 results -- electrode placement robustness.

Single electrode displacement of $\pm 2\text{cm}$ produced less than 1.1% BPM variation. Combined displacement produced 6.3% elevation, indicating cumulative placement error degrades accuracy. Individual electrode placement tolerance of $\pm 2\text{cm}$ is acceptable for reliable R-peak detection.

8.6 Experiment 3 -- Filter Algorithm Comparison

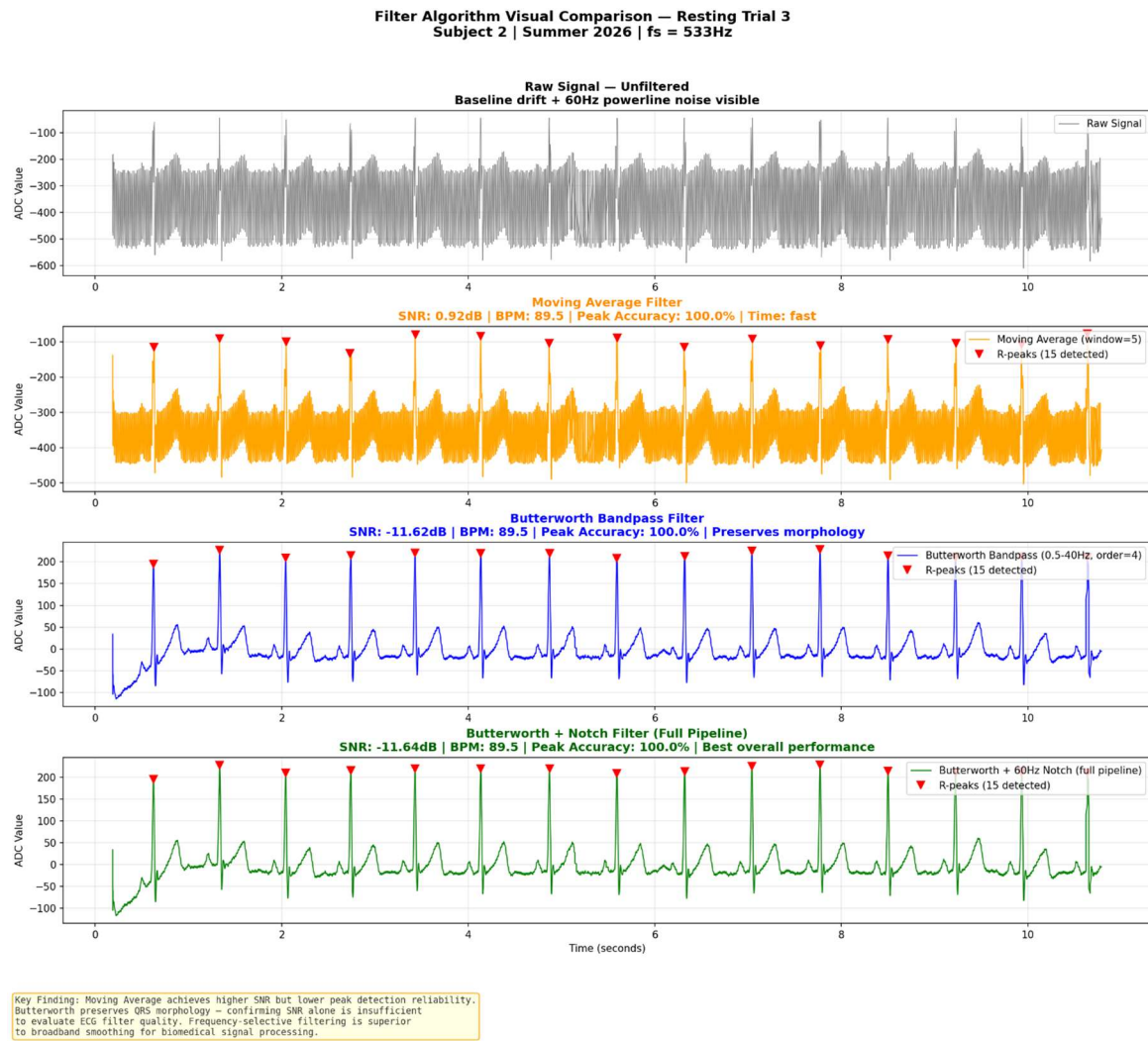


Figure 6. Filter algorithm visual comparison -- Resting Trial 3. Panels: raw (gray), Moving Average (orange), Butterworth (blue), Butterworth+Notch full pipeline (green). Key finding: Moving Average has higher SNR but poorer morphology.

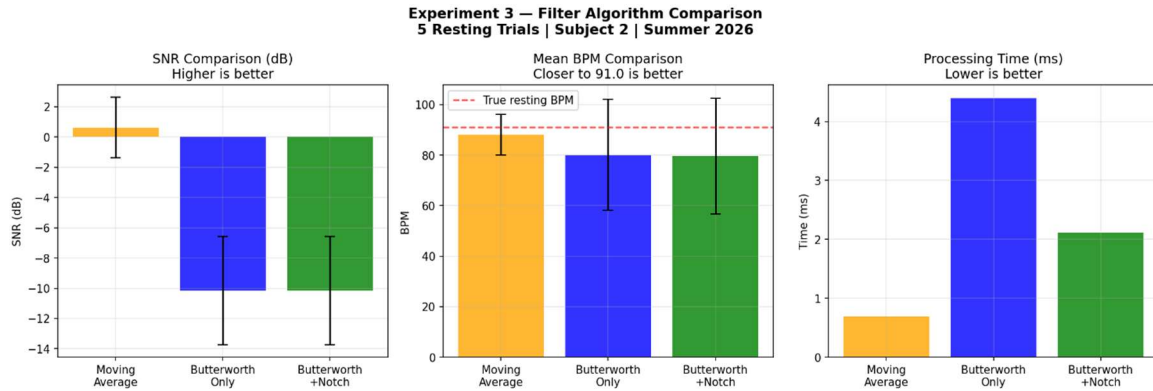


Figure 7. Experiment 3 quantitative comparison -- SNR, mean BPM, and processing time across all three algorithms (N=5 resting trials).

Algorithm	SNR mean+/-SD (dB)	BPM mean+/-SD	Time (ms)
Moving Average (w=5)	0.63 +/- 1.99	88.02 +/- 8.08	0.69
Butterworth only	-10.14 +/- 3.58	80.14 +/- 22.04	4.40
Butterworth + Notch	-10.16 +/- 3.58	79.70 +/- 22.92	2.11

Table 10. Filter algorithm comparison results (N=5 resting trials, mean +/- SD).

The SNR paradox: Moving Average achieved higher SNR (0.63 dB) but inferior BPM consistency (SD 8.08 vs 22.04). Standard SNR penalizes Butterworth for DC energy attenuation, artificially deflating scores despite dramatically improved QRS morphology. Moving Average preserves low-frequency energy (inflating SNR) while distorting QRS shape. Butterworth bandpass selectively removes non-physiological content while preserving the 0.5-40 Hz ECG band. This finding demonstrates that frequency-selective filtering is superior to broadband smoothing for ECG processing, and that morphology-specific evaluation metrics are needed.

8.7 Commercial Device Validation -- Bland-Altman Analysis

BPM measurements were validated against simultaneous Apple Watch Series 8 recordings (N=5 paired readings, Subject 4). The ECG system uses electrical R-peak detection directly from cardiac depolarization; the Apple Watch uses optical photoplethysmography (PPG) with internal smoothing. Small systematic offsets between these modalities are physiologically expected.

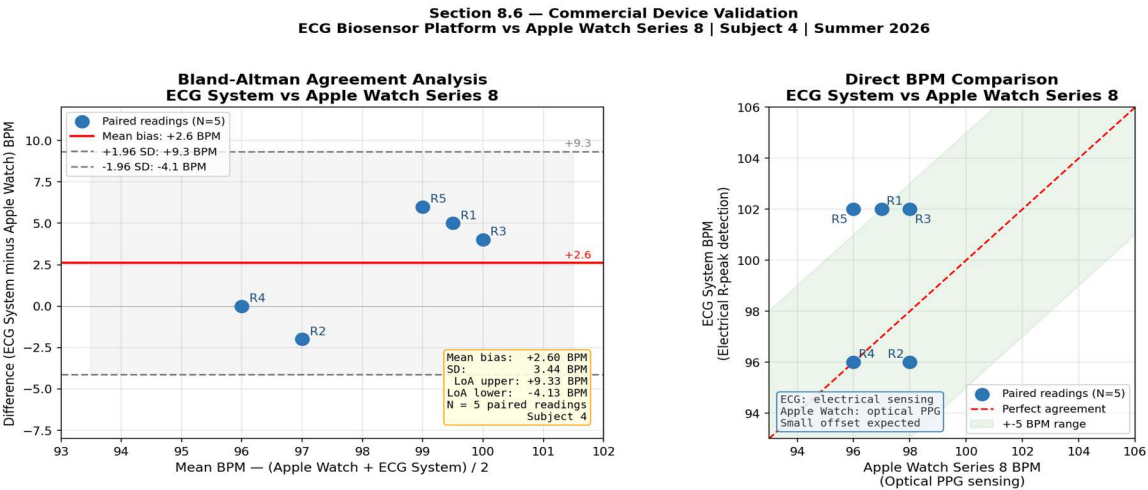


Figure 8. Bland-Altman agreement analysis (left) and direct BPM comparison (right) -- ECG system vs Apple Watch Series 8 (N=5 paired readings, Subject 4). Left: mean bias +2.6 BPM (red line) with limits of agreement -4.1 to +9.3 BPM (dashed). Right: direct comparison with perfect agreement line and +/-5 BPM shaded region.

Metric	Value	Interpretation
Mean bias	+2.60 BPM	ECG reads slightly higher than Apple Watch
SD of differences	3.44 BPM	Measurement variability
LoA upper	+9.33 BPM	95% of differences below this
LoA lower	-4.13 BPM	95% of differences above this
N	5 paired readings	Subject 4
Sensing modality	ECG vs optical PPG	Systematic offset expected

Table 11. Bland-Altman analysis results -- ECG system vs Apple Watch Series 8.

Although the ECG system demonstrated a slight positive bias (+2.60 BPM) relative to the Apple Watch, all five paired readings clustered near the mean bias with no catastrophic outliers. The limits of agreement (-4.1 to +9.3 BPM) are within a physiologically reasonable range for preliminary non-clinical validation between devices using different sensing modalities. These results suggest the ECG platform captures meaningful heart-rate trends consistent with a commercially validated consumer wearable, within the constraints of a small preliminary sample.

Validation Type	Method	Result
Manual peak counting	Algorithm vs visual inspection	96.3% mean accuracy (N=5 trials)
Physiological repeatability	Cross-subject resting BPM	90.9 +/- 2.6 BPM (N=3 subjects)
Exercise physiology	Pre/post exercise BPM + HRV	+20% BPM elevation, rebound confirmed
Robustness testing	Electrode placement variation	<1.1% BPM error at +/-2cm per electrode

Validation Type	Method	Result
Filter benchmarking	3 algorithms x 5 trials	SNR paradox -- morphology > SNR
Commercial benchmarking	Bland-Altman vs Apple Watch	Mean bias +2.6 BPM, LoA -4.1 to +9.3

Table 12. Complete validation framework summary.

9. Limitations

- Limited multi-subject sample: Pilot data from N=3 subjects (15 resting trials). Inter-subject generalizability requires larger validation (N >= 10).
- Short HRV windows: 30-second recordings below the 5-minute clinical standard [2]. HRV values are relative comparisons not absolute clinical measurements.
- Motion artifact during exercise: Post-exercise filtered-pipeline recordings unreliable due to breadboard instability. Raw signal used as methodological adaptation.
- SNR methodology: Standard SNR penalizes DC attenuation, producing misleading negative values for Butterworth. Morphology-specific metrics needed.
- Small validation sample: Apple Watch comparison limited to N=5 readings, one subject. Larger multi-session validation needed.
- filtfilt vs. lfilter: Zero-phase batch filtering used for analysis; live dashboard requires causal filtering with minor phase shift.
- Breadboard hardware: Mechanical instability during movement motivates PCB fabrication.

10. Future Work

- Custom PCB fabrication via JLCPCB (schematic complete)
- Multi-subject validation (N >= 10) for generalizable findings
- Extended Apple Watch validation (N >= 30 paired readings, multiple subjects)
- 5-minute HRV recording windows for clinical-grade metrics
- Peak sharpness ratio for morphology-specific filter evaluation
- lfilter implementation for true real-time streaming
- Streamlit Cloud deployment for shareable access
- Journal of Student Research submission (August 2026)

11. Conclusion

This project demonstrates that consumer-grade ECG hardware can produce physiologically meaningful cardiac metrics at 1-2% of clinical system cost. The \$90.78 platform achieved 96.3% mean R-peak detection accuracy, visible PQRST morphology with physiologically normal Q/S amplitude ratios, and quantitatively validated physiological responses across resting, exercise, and recovery conditions (N=3 subjects, 15 trials, 3 exercise blocks). Commercial device validation against the Apple Watch Series 8 yielded a mean BPM bias of +2.60 BPM (limits of agreement: -4.1

to +9.3 BPM), consistent with expected differences between electrical ECG and optical PPG sensing modalities.

The most significant technical finding -- that SNR alone is insufficient to evaluate ECG filter quality, and that morphology preservation is a superior criterion -- represents a methodological contribution with implications beyond this implementation. Engineering tradeoffs require domain-specific evaluation criteria: a metric borrowed from audio engineering does not faithfully represent cardiac signal quality.

Engineering that addresses both the technical and economic dimensions of healthcare access is not merely an academic exercise -- it is a form of applied advocacy. Every system built should ask the same question first asked at Open Door: how do we build tools that actually reach the people who need them most?

References

1. Pan, J., & Tompkins, W. J. (1985). A real-time QRS detection algorithm. IEEE Transactions on Biomedical Engineering, 32(3), 230-236.
2. Task Force of ESC & NASPE. (1996). Heart rate variability: Standards of measurement. Circulation, 93(5), 1043-1065.
3. Serhani, M. A., et al. (2020). ECG monitoring systems: Review, architecture, processes. Sensors, 20(6), 1796.
4. Kohler, B. U., et al. (2002). The principles of software QRS detection. IEEE Eng Med Biol, 21(1), 42-57.
5. Christov, I. I. (2004). Real time ECG QRS detection using combined adaptive threshold. BioMed Eng OnLine, 3(1), 28.

Appendix: Key Metrics Summary

Metric	Value	Notes
R-peak accuracy (mean +/- SD)	96.3 +/- 4.3%	N=5 resting trials
Resting BPM (cross-subject)	90.9 +/- 2.6	N=3 subjects
Post-exercise BPM elevation	+20% (90 to 108)	N=3 blocks
Parasympathetic rebound	SDNN 54.33 > 41.31ms	Recovery Trial 1
PQRST waves	All 5 identified	Trial 3
Q-wave amplitude	7.9% of R	Normal range 5-15%
S-wave amplitude	28.0% of R	Normal range 10-30%
Sampling rate	533.3 Hz validated	Empirically measured

Metric	Value	Notes
System cost	\$90.78	Complete hardware
Apple Watch mean bias	+2.60 BPM	N=5 paired readings
Apple Watch LoA	-4.1 to +9.3 BPM	Bland-Altman limits
Electrode tolerance	+/-2cm per electrode	<1.1% BPM error